Appendix C

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1. Let X be a disjoint union of locally closed subschemes X_i. Then show that

$$Z(X,t) = \prod Z(X_i,t).$$

2. Let $X = \mathbb{P}^n_k$, where $k = \mathbb{F}_q$, and show from the definition of the zeta function that

$$Z(\mathbb{P}^n_k;t) = \frac{1}{(1-t)(1-qt)\cdots(1-q^nt)}.$$

Verify the Weil conjectures for $\mathbb{P}^n_k$.

Proof. We can write $\mathbb{P}^n_k$ as a union of locally closed subschemes

$$\mathbb{P}^n_k = \mathbb{A}^n_k \cup \mathbb{A}^{n-1}_k \cup \cdots \cup \mathbb{A}^1_k \cup \mathbb{A}^0_k,$$

where we can take $\mathbb{A}^0_k = \operatorname{Spec} k$ to have $\mathbb{F}_q$-coordinates. Over any finite field and $n \geq 0$, $\mathbb{A}^n_k$ has q^n number of $\mathbb{F}_q$-points. Hence, $N_r(\mathbb{A}^n_k) = q^{rn}$, and it is easy to compute

$$Z(\mathbb{A}^n_k,t) = \exp\left(\sum_{r=1}^{\infty} N_r(\mathbb{A}^n_k)\frac{t^r}{r}\right) = \exp\left(\sum_{r=1}^{\infty}\frac{(q^nt)^r}{r}\right) = \frac{1}{1-q^nt}.$$

Combining with (Ex. 1), we have

$$Z(\mathbb{P}^n_k;t) = \prod_{i=0}^{n} Z(\mathbb{A}^i_k;t) = \frac{1}{(1-t)(1-qt)\cdots(1-q^nt)}.$$

□

3. Let X be a scheme of finite type over $\mathbb{F}_q$, and let $\mathbb{A}^1$ be th affine line. Show that

$$Z(X \times \mathbb{A}^1, t) = Z(X, qt).$$

4. The *Riemann zeta function* is defined as

$$\zeta(s) = \prod \frac{1}{1-p^{-s}},$$

for $s \in \boldsymbol{C}$, the product being taken over all prime integers p. If we regard this function as being associated with the scheme $\operatorname{Spec}\mathbb{Z}$, it is natural to define, for any scheme X of finite type over $\operatorname{Spec}\mathbb{Z}$,

$$\zeta_X(s) = \prod (1 - N(x)^{-s})^{-1}$$

where the product is taken over all closed points $x \in X$, and $N(x)$ denotes the number of elements in the residue field $k(x)$. Show that if X is of finite type over $\mathbb{F}_q$, then this function is connected to $Z(X,t)$ by the formula

$$\zeta_X(s) = Z(X, q^{-s}).$$

5. Let X be a curve of genus g over k. Assuming the statements (1.1) to (1.4) of the Weil conjectures, show that $N_1, N_2, \ldots, N_g$ determine N_r for all $r \geq 1$.

6. Use (IV, Ex. 4.16) to prove the Weil conjectures for elliptic curves. First note that for any r,

$$N_r = q^r - (f^r + \hat{f}^r) + 1,$$

where $f = F'$. Then calculate $Z(t)$ formally and conclude that

$$Z(t) = \frac{(1 - ft)(1 - \hat{f}t)}{(1 - t)(1 - qt)}$$

and hence

$$Z(t) = \frac{1 - at + qt^2}{(1 - t)(1 - qt)},$$

where $f + \hat{f} = a_X$. This proves rationality immediately. Verify the functional equation. Finally, if we write

$$1 - at + qt^2 = (1 - \alpha t)(1 - \beta t),$$

show that $|a| \leq 2\sqrt{q}$ if and only if $|\alpha| = |\beta| = \sqrt{q}$. Thus the analogue of the Riemann hypothesis is just (IV, Ex. 4.16d).

7. Use (V, Ex. 1.10) to prove the analogue of the Riemann hypothesis (1.3) for any curve C of genus g defined over $\mathbb{F}_q$. Write $N_r = 1 - a_r + q^r$. Then according to (V, Ex. 1.10),

$$|a_r| \leq 2g\sqrt{q^r}.$$

On the other hand, by (4.2) the zeta function of C can be written

$$Z(t) = \frac{P_1(t)}{(1 - t)(1 - qt)}$$

where

$$P_1(t) = \prod_{i=1}^{2g}(1 - \alpha_i t)$$

is a polynomial of degree $2g = \dim \mathrm{H}^1(C, \mathbb{Q}_\ell)$.

(a) Using the definition of the zeta function and taking logs, show that

$$a_r = \sum_{i=1}^{2g}(\alpha_i)^r$$

for each r.

(b) Next show that $|a_r| \leq 2g\sqrt{q^r}$ for all $r \iff |\alpha_i| \leq \sqrt{q}$ for all i.

(c) Finally, use the functional equation (4.4) to show that $|\alpha_i| \leq \sqrt{q}$ for all i implies that $|\alpha_i| = \sqrt{q}$ for all i.